
Validation Windows: Epistemic Uncertainty Produced by the Solver in a Literature-Derived Bayesian Network

Deborah Vakas Duong¹ Igor Yi¹

Abstract

When a Bayesian network is built from published scientific studies, each contributing a point estimate and a 95% confidence interval, the aggregation problem is fundamentally epistemic, not just statistical. Different studies on the same edge disagree; different edges imply joint probabilities that must satisfy the law of total probability; and the resulting conditional probability tables (CPTs) must also be consistent with observed population frequencies. We describe a system that solves the aggregation as a quadratic program whose feasible region is a convex polytope of CPTs consistent with all sources of evidence simultaneously. The width of the polytope along each study’s axis, which we call the validation window, is a first-class representation of epistemic uncertainty for that edge: produced by the construction itself, not computed post-hoc as a calibration layer. It is tight where literature and population data agree, wide where they disagree, and exceeds the study’s own literature CI when the solver must stretch the study to achieve global consistency. The system sits between two transformer endpoints: an upstream LLM that extracts literature into structured spreadsheet rows (making the pipeline domain-portable), and a downstream multi-agent coevolutionary simulation it seeds with behaviors. On a 427-node longevity-risk network built from 166 published studies (72% meta-analyses, $\sim 4 \times 10^8$ pooled subjects), the median window shrinks by 87% (from 0.125 to 0.016)

as consistent literature-established edges are added. A residual set of low-base-rate edges where the solver cannot commit to a direction, the $\approx 2 \times 10^{-4}$ pancreatic-cancer chain, demonstrates the intended failure mode: abstention, visible as inversion of the CPT direction precisely where literature alone cannot discipline the joint. Beyond the window, the construction is also usable as a predictor: on a post-submission improved build, with each target’s definitional-surrogate biomarkers excluded, it discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.73 across 15 constructed disease targets (up to 0.94 on coronary artery disease), at or above the level of cohort-fitted clinical risk calculators on the better-instrumented targets. The window thus provides a direct, per-edge measure of what the model does not know, computed by the construction itself.

1. Introduction

A Bayesian network built from the published scientific literature faces three simultaneous epistemic challenges. Individual uncertainty: each primary or meta-analytic study reports a point estimate and an interval, not a single exact value. Cross-study disagreement: two studies on the same edge often report different effect sizes with overlapping but non-identical intervals. Joint coherence: a collection of marginal effect sizes, assembled into a network, must respect the law of total probability and must remain consistent with population-level frequencies from a reference survey dataset. Collapsing all of this to a single CPT per node, the default in published expert systems (Heckerman, 1998), discards most of the epistemic information before it reaches the user. Returning the set of CPTs consistent with all sources, by contrast, preserves the shape of what is not known.

Transformers sit on both ends of the pipeline we de-

¹Agent Based Learning Systems, San Luis Obispo, California, USA. Correspondence to: Deborah Vakas Duong <dduong@agentBasedLearningSystems.com>.

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scribe. Upstream, a large language model (Brown et al., 2020) reads published meta-analyses and primary studies, extracting effect sizes, confidence intervals, populations, and citations into the structured spreadsheet rows our solver consumes. The LLM also supplies the causal structure the survey cannot: edge directions and mediating-node placement, from the literature’s mechanistic consensus (App. D). This LLM-mediated extraction is what makes the method generalizable: any scientific domain with peer-reviewed effect-size literature, longevity and health in our demonstration, but also macroeconomic policy, education interventions, or criminal-justice evaluation, can be converted to a network without domain-specific hand-curation. Downstream, once the network carries treatment (interventional) edges it is itself a policy net that seeds the agents of a multi-agent coevolutionary simulation with behaviors.

This paper presents the network-construction step. The system takes as input a spreadsheet of literature-sourced edges (effect sizes plus 95% confidence intervals) together with a reference dataset of population covariate frequencies (we use NHANES (Centers for Disease Control and Prevention, National Center for Health Statistics, 2020) throughout), and produces a Bayesian network whose CPTs are the solution of a quadratic program. The QP’s objective is heuristic, a least-squares fit to the literature point estimates plus a monotonicity penalty that discourages direction inversions, and its constraints are probability-law: the CPT is a valid probability table, each literature RR lies within a windowed band, and the law of total probability holds across groups of cells. We call the last item the subset constraints. The feasible region of the constrained QP is a convex polytope of CPTs consistent with all sources of evidence simultaneously.

Our contribution is conceptual and algorithmic: a working quadratic-program solver that constructs the polytope from literature in a single pass, together with the proof (Theorem 3) and argument that the width of the feasible polytope along each study’s axis, the validation window, is the quantity the system already computes that directly answers the core epistemic-uncertainty question: for this edge, how much does the model not know? Tight windows mean “literature and population data mutually confirm this effect under the structure I’ve built.” Wide windows mean “I had to bend the literature to make the whole thing fit.” Windows that exceed the raw literature CI mean “my structure may be wrong; attend here.” The system’s operating loop iteratively searches for structural changes or additional literature that narrows the widest windows; the collective width of the polytope

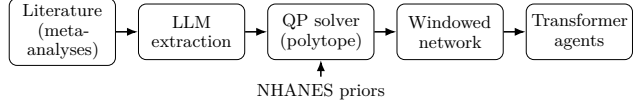


Figure 1. The end-to-end pipeline. An LLM extracts published effect sizes into structured rows; the QP solver returns the feasible polytope of CPTs; each edge carries its validation window into the downstream transformer agents. NHANES supplies per-node priors to the solver.

serves as the monotone-decreasing measure of residual epistemic uncertainty. When the network is handed off to downstream transformer agents, these windows travel with each edge: the transformer receives a prior that makes its own uncertainty commensurate with the literature’s. As supplementary evidence that these literature-constrained CPTs carry real predictive structure, an improved post-submission build of the network discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.73 (up to 0.94), with diagnostic biomarkers excluded (App. J).

The system is not an imprecise-probability engine (§3 relates the feasible polytope to credal sets (Walley, 1991; Cozman, 2000)). It is not a calibration (Guo et al., 2017) or conformal-prediction (Vovk et al., 2005) overlay on a pre-trained classifier. It is a construction process whose intermediate artifact, the windowed polytope, is the epistemic representation. The construction itself reports the residual uncertainty on each edge.

2. The System End-to-End

Figure 1 summarises the pipeline end to end; the three construction stages (Inputs, Compilation, CPT solve) are described in turn.

Inputs. A single spreadsheet. A study row carries an output node (e.g. lung_cancer), an input node and its value (smoking_yes), a statistic type (RR, OR, HR, or effect size), a point estimate, a $\pm$ for the 95% CI, and a citation; our demonstration network has 166 such rows, 72% of which are meta-analyses, with a pooled total of $\sim 4.17 \times 10^8$ subjects. A definition row declares a node’s type: a directly-observed NHANES variable, a dependency node whose CPT the QP solves from its parents’ study RRs, or a logical aggregator (any_of, all_of) combining parents. Still other rows declare a link between nodes: a subset relation (is_a, subsumes) or identity (equivalent_to) for definitional containment, or a distal link routing a computed aggregator-level effect onto a disease. The build also takes the reference dataset NHANES 2017–March 2020, 116K re-

spondents.

Compilation. The spreadsheet compiles to a DAG with one node per output and one edge per (input, output) pair. Aggregator definitions resolve to deterministic truth tables. Distal links, 33 in our demonstration, organizing leaves into sub-aggregators that feed a disease, are handled by a Monte-Carlo procedure that propagates study-level RRs through the aggregator’s NHANES-weighted co-occurrence structure to produce a single aggregator-to-disease RR with its own 95% CI.

CPT solve. For each node Y with parents $X_1 \dots X_K$, let $\mathbf{x}$ be the flattened CPT. Each literature study i bearing on Y gives four linear constraints $r_k^{(i)} \mathbf{x} = s_k^{(i)}$ (the standard marginal-RR expansion): each $r_k^{(i)}$ expresses study i ’s marginal RR as a linear functional of the flattened CPT via the law of total probability, fixed by the study’s (parent, value, target) and the reference-survey marginal, not a free choice. Stacking these rows gives the matrix R (the collection of the $r_k^{(i)}$) and target vector $\mathbf{s}$. The QP objective (both terms heuristic) is

$$\mathcal{L}(\mathbf{x}) = \|\mathbf{R}\mathbf{x} - \mathbf{s}\|_2^2 + \lambda \sum_p \max(0, \text{inv}_p)^2, \quad (1)$$

where the first term fits the literature point estimates and the second penalises monotonicity inversions, with weight $\lambda > 0$ (p indexes signed-direction (parent, value) pairs and inv_p is the violation amount, zero when the direction holds). We do not claim this objective follows from a Bayesian or decision-theoretic principle; it is a simple heuristic that merely selects one interior point of the polytope. The probability laws live in the constraints, not the objective, so the reported window W depends only on the constraint polytope (§3, App. A) and is invariant to λ . The constraints that encode probability laws are: (i) $\mathbf{x}$ is a valid probability table (row sums, non-negativity); (ii) a windowed literature band per study, $s - c \leq \mathbf{R}\mathbf{x} \leq s + c$, with c scaled per-study by the study’s own literature-CI width (narrower CIs yield tighter effective bounds) and the global scale determined by a binary search on feasibility; (iii) subset constraints, a set of linear equalities/inequalities requiring that, for every group of CPT cells linked by the law of total probability, the corresponding relation holds within the windowed tolerance. Subset constraints range over groups of cells induced by the CPT’s full structure, including marginalisation over any subset of parents. Grouping the terms this way isolates the heuristic part of the solve (literature-fit and monotonicity) from the probability-law part

(the constraints). Least squares and monotonicity are retained as soft heuristics because marginal effect sizes and their directions are known properties of the literature, not probability laws. NHANES supplies per-node priors through the definition rows; it does not supply conditional training signal to the solver, which would create a circularity in the evaluation metrics below.

The validation window W is a confidence interval. The polytope’s boundary is determined by the probability-law constraints, not by the objective; W is therefore independent of λ . The window c is not a free parameter, it is determined by a binary search on feasibility. We start with $c=1.0$, which trivially admits any probability table, and shrink c toward 0 as far as the constrained QP remains feasible. The smallest feasible c is recorded per edge as the edge’s validation window, $W \in [0, 1]$, in the same probability units as the CPT cells it bounds. $W=0$ means the constraints uniquely determine the edge’s CPT row; $W=1$ means the constraints are non-binding.

Under the normal approximation, each input 95% CI is a likelihood-ratio confidence region at level $\alpha=0.05$, so W_e is the half-width of a confidence interval for the edge’s CPT cell and the solver reports a per-edge CI coverage cov_e alongside W_e . Coverage scales with the input-band factor c via $c \cdot z_{1-\alpha/2} = z_{1-\alpha/(2n)}$ (Bonferroni); for $n=166$ studies, joint coverage 0.95 corresponds to $c \approx 1.85$. Runs in this paper use $c=1$; intervals are exact marginally and conservative jointly, and can be re-generated at any target joint coverage via the Bonferroni relation. When studies on different edges are mutually consistent, W collapses toward the literature CI width; when they are not, c must widen past the literature CI to avoid infeasibility, and the amount by which W exceeds the literature CI is the system’s explicit complaint.

3. Windows as Epistemic Uncertainty

The feasible polytope of CPTs satisfying all constraints at the minimum-feasible c is a convex region (Fig. 2). Along each study axis its extent is $[L_i, U_i]$, and $W_i = U_i - L_i \in [0, 1]$ is the validation window for that study. By construction this is a projected credal set in the sense of Walley (1991) and Cozman (2000), but arrived at constructively from published literature rather than from elicited lower and upper probabilities. Four properties make W a useful epistemic report:

- (1) Monotone in evidence. Adding a consistent literature edge can only tighten the polytope (Theorem 3, part 1), and a binding one strictly tightens it (part 2);

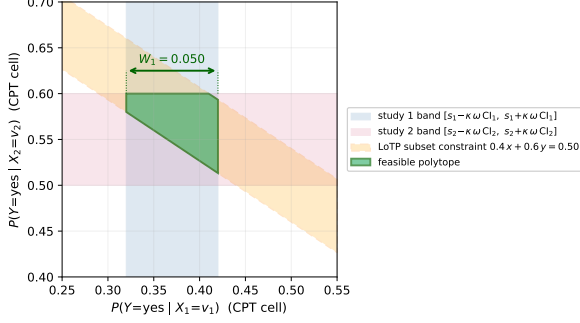


Figure 2. Feasible polytope of CPTs from two studies and a law-of-total-probability (LoTP) subset constraint. Axes are CPT-cell probability units, e.g. $P(Y=\text{yes} \mid X_1=v_1)$ on the horizontal axis (each study’s literature RR multiplied by the base rate gives a band on its corresponding axis). Blue and pink: the two literature CI bands. Orange dashed: the LoTP subset-constraint band. Green polygon: the feasible polytope (solution set); its projection onto the study-1 axis is the feasibility band W_1 . When the projection equals the literature CI the system has extracted no extra information; when narrower, cross-CPT coherence has tightened the prior; when wider, the solver has stretched the study to keep the QP feasible, signalling cross-study tension.

an inconsistent edge is detected because c must widen (or the QP becomes infeasible). In our experiments the median window $\tilde{W}$ dropped by 87% (from 0.125 to 0.016) as eight structural edges from a medical-knowledge audit were added, meaning that by the final build, half of all solved edges have $W \leq 0.016$ (their literature + subset-constraint content is almost fully binding), and the remaining half spread up to 1.0.

(2) Localizes cross-edge tension. The system sorts studies by W . Studies with high W are edges where the per-study literature CI disagrees with the joint of all other constraints in the network: either literature on this edge contradicts literature on an adjacent edge, or literature contradicts population data, or the DAG is missing a structural edge that would resolve the tension. In the demonstration network the top-20 window list after each iteration provided the to-do list for the next iteration. The three edges that remain stuck at $W=1$ after all available evidence has been added are the diabetes/healthy-diet pairs; their literature pulls in a direction that no current structural change has reconciled with the adjacent constraints.

(3) Exceeds the literature CI when forced. A window wider than the study’s own literature CI is a report by the system that something upstream is wrong: the chosen structure, another study, or a NHANES leaf mapping.

(4) Determines untested parent combinations. For K parents the CPT has 2^K rows; literature typically measures only the K marginal effects. Given the marginals and the reference survey’s parent co-occurrence structure, the subset constraints mathematically determine the remaining $2^K - K$ rows to within the window, combinations no study has jointly examined, provided the co-occurrence exceeds a threshold we make precise in App. C. We measure co-occurrence with a signed deviation, $\rho(X_i, X_j) = \max_{v_i} [P(X_j = \hat{x}_j \mid X_i = v_i) - P(X_j = \hat{x}_j)]$, with $\hat{x}_j$ the parent’s reference state and the sign of the maximising deviation retained. Because the derivation is the law of total probability, these predictions need no further experiment. The required $|\rho|$ is modest; where it is not yet sufficient, the objective returns the best estimate given the known marginals and the DAG, with the window reporting the residual uncertainty.

A complementary abstention signature (§5) is direction symmetry within a small feasible polytope: W can be modest yet the solver cannot pick a committed direction. That failure does not surface in W ; it surfaces in the query-RR direction check.

4. Empirical Demonstration

Setup. Our demonstration network covers longevity risk: 427 nodes (227 epidemiological variables from NHANES plus 200 structural nodes including 15 disease targets, 33 distal aggregators, and 152 definitional sub-variables); 166 study rows (72% meta-analyses, $\sim 4.17 \times 10^8$ pooled subjects); a 116K-respondent NHANES reference set. We evaluate three things per build:

- Direction accuracy: for each cited edge, clamp its input at yes, query its target, compare the sign of the resulting query RR to the literature RR’s sign; report the fraction correct out of 369 testable cited edges.
- Median window $\tilde{W}$: median of W_i across all 287 nodes whose CPTs are solved.
- Joint fidelity within 5%: exhaustive over every CPT cell for which a NHANES ground truth exists, i.e., every parent of the node and the node itself carry NHANES codes, and ≥ 30 NHANES respondents match the parent combination. Latent aggregators and distal nodes are excluded only because they are not directly observed (no ground truth), not by selection. Under this rule the demonstration network yields 178 cells covering 15 nodes, rising to 194 cells / 16 nodes after the final evidence-addition row (which introduces a new all-NHANES-observable node). We report the fraction with

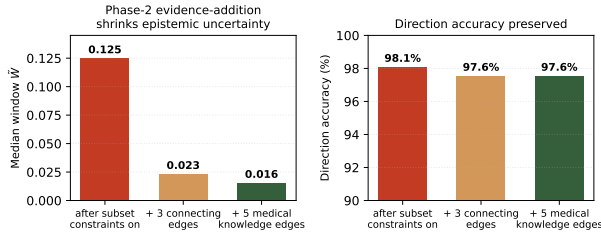


Figure 3. Window-reduction and direction-accuracy progression as consistent literature edges are added. Direction accuracy (right) holds within 0.5 percentage points while median window (left) shrinks 87%. Data from Table 1.

$$|\text{CPT} - \text{NHANES-empirical}| < 0.05.$$

Cycles of evidence addition. Table 1 and Fig. 3 show median window $\tilde{W}$ and direction accuracy as structural edges from a medical-knowledge audit are added. After eight subset-on iterations beginning with the 166-study network, $\tilde{W}$ collapses from 0.125 to 0.016, an 87% reduction in epistemic slack, while the direction accuracy remains above 97.5%. The structural-audit edges (including BMI→diabetes, diabetes→CAD, BMI→hypertension, sleep-apnea→hypertension, alcohol→liver cancer, frailty→all-cause mortality, depression→CAD) are each supported by published meta-analyses; they tighten the polytope because they supply the cross-edge coherence that resolves previously-unresolved tensions in the network.

The widest- W studies as research agenda. In the final build the twenty widest- W studies cluster into three groups: (a) transportability-limited studies whose exposure definitions do not map cleanly onto NHANES covariates (occupational nickel exposure vs. NHANES urinary nickel); (b) structurally underspecified chains where the disease has few parents and the solver has latitude; (c) correlated co-parents entangled in NHANES but represented as independent in the network. Each group suggests a distinct next-action: find a population-based study, add a directly-observed biomarker, or introduce a dependency constraint between the co-parents. The widest- W list is the system’s internally-generated research agenda, deriving next actions from explicit ignorance localization, rather than from a flat confidence score.

5. Abstention at Low Base Rate

Pancreatic cancer has a NHANES prevalence of $\approx 2 \times 10^{-4}$. Seven studies linking occupational chemical exposures to pancreatic cancer, with literature RRs

1.3–1.7, do not produce the expected direction-correct CPT cells in our solved network. The distal-computed RR (aggregator → disease) is 2.60; but the QP’s solution assigns $P(\text{panc}|\text{chem}=\text{yes}) < P(\text{panc}|\text{chem}=\text{no})$, the opposite direction.

This is not a bug. At base rate 2×10^{-4} the absolute-error landscape of the LS term is vanishing (10^{-7} units), and the monotonicity penalty at the default margin tolerates inversions of magnitude 10^{-2} without penalty. With both soft objective terms effectively flat, the solver’s choice is driven by the hard subset constraints alone, and at such low marginal, the law-of-total-probability relations over the CPT cells admit a feasible region nearly symmetric along the parent-direction axis. Any point in that region satisfies the constraints equally well; the solver’s converged point falls on the direction-inverted side. This is the correct report that literature plus the law of total probability alone do not uniquely determine the conditional direction at this base rate. In the system’s own language it is reporting that it abstains.

The per-study window W does not catch this case: in cycle-12 data the seven pancreatic studies have windows in the mid-range (not in the top-20 widest), because the solver did find a feasible point within each literature CI, it just happened to pick one on the direction-inverted side of a symmetric feasible region. W reports the radius of the feasible region; it does not report the symmetry of that region under reflection through the direction-neutral point. The query-RR direction check (§4) catches it: the seven pancreatic edges all return query RR = 0.055 against literature RR 1.3–1.7, a maximum-signal disagreement. A deployment layer should therefore flag any query whose RR reverses the literature sign as a potential abstention, independent of W , and route the user to the widest- W and direction-inverted studies together with the statement “at this base rate, available literature does not uniquely determine the conditional direction.”

We attempted two solver-side remediations, scaling the monotonicity-penalty weight by $10 \times$ and tightening the inversion tolerance from 0.01 to 0.0001, and neither dislodged the inversion, confirming that the issue is geometric (the feasible polytope at low base rate is symmetric along the direction axis) and not a matter of tuning. The right treatment is post-solver: at query time, when a queried cell lies in a region of the polytope whose symmetric reflection is also feasible, the response should be the window rather than the point. This is a lifelong-learning operation: when new literature or population data narrow the polytope, the cell resolves; until then, the system knows it does not

Table 1. Evidence-addition iterations. Subset constraints are on throughout. Adding literature-established edges (rows 2–3) reduces $\bar{W}$ by 87% while holding direction accuracy within 0.5 percentage points; joint fidelity rises from 77.0% to 78.9%. The “Median CI coverage” column is $\text{erf}(W \cdot z_{0.975}/\sqrt{2})$, the probability a single-study 95%-CI-derived band of that width would contain the true value; scaling inputs by the Bonferroni factor $c \approx 1.85$ lifts every coverage to joint-95%. A CI coverage level, not an NHST p -value.

Build	Direction accuracy	Median $\bar{W}$	Median CI coverage	Joint within 5%
Subset constraints on, 166 studies	362/369 (98.1%)	0.125	0.194	137/178 (77.0%)
+ 3 connecting-study edges (e.g. bmi→diabetes, diabetes→CAD)	360/369 (97.6%)	0.023	0.036	137/178 (77.0%)
+ 5 medical-knowledge edges (bmi→hypertension, etc.)	360/369 (97.6%)	0.016	0.025	153/194 (78.9%)

know.

6. Discussion

Although the window is the paper’s contribution, the constructed network is also usable as a predictor: on a post-submission improved build it discriminates held-out clinical events at a mean ROC AUC of 0.73 across 15 targets (up to 0.94, biomarkers excluded; App. J), supplementary evidence that the literature-constrained CPTs carry real predictive structure, not a contribution of the method itself.

Each study row is an update operator: consistent evidence shrinks the polytope and lowers W , while disagreement forces c wider and surfaces in the widest- W list, so re-solving on every update keeps W a faithful measure of residual disagreement. This affords two immediate uses, automatic curation (accept a row if the median window decreases, flag it otherwise) and hypothesis generation (rank candidate structural changes by polytope narrowing), and the crowdsourced construction loop of App. N. The spreadsheet rows are LLM-extracted (Brown et al., 2020) following a released construction manual, making the method domain-portable, with the LLM’s errors made auditable as wide windows (App. F). The motivating downstream use is to seed multi-agent simulations from the windowed polytope, so agents disagree in proportion to literature disagreement.

7. Related Work

Imprecise probability (Walley, 1991) and credal networks (Cozman, 2000; Piatti et al., 2010) formalise uncertainty as a convex set of distributions, with LP-based inference algorithms developed by Antonucci et al. (2015) and surveyed by Mauá and Cozman (2020); recent work carries credal sets into deep uncertainty quantification (Caprio et al., 2024). Our feasible polytope is isomorphic to a separately specified local credal set, but constructed from published liter-

ature rather than from elicited lower–upper bounds; the convex set arises as a by-product of a literature-constrained construction rather than being learned or elicited. Dirichlet distributions over CPT rows (Heckerman, 1998) do not couple across edges; our subset constraints do. Calibration (Guo et al., 2017; Kuleshov et al., 2018) and conformal prediction (Vovk et al., 2005; Romano et al., 2019) adjust outputs post hoc; we report the solver’s own feasibility interval. Classifier abstention (Hendrycks and Gimpel, 2017; Liu et al., 2020) declines to predict; we abstain where the polytope is symmetric along the direction axis. Retrieval-augmented generation (Lewis et al., 2020) uses literature at inference; we use literature as hard constraints at construction.

The causal-direction layer parallels expert-elicited DAG construction in epidemiological causal inference (Hernán and Robins, 2020), the LLM serving as the domain expert; its dependence- and mediation-handling (App. D) are construction-vocabulary analogues of operations studied by Pearl (1988) and VanderWeele (2015), yielding literature-coherent observational conditionals informed by causal direction, not a claim of Pearl-sense interventional identification.

8. Conclusion

The validation window W is produced by the model-construction procedure itself. On the 427-node network the polytope shrank 87% as evidence was added, now provably monotone (Theorem 3); the widest windows named research directions. Where the solver could not commit (the pancreatic-cancer chain at 2×10^{-4}) it abstained via a direction-symmetric polytope, operationalising a model that knows what it does not know.

Code and data. <https://github.com/agentbasedlearningsystems/validation-windows> reproduces the headline results.

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Appendix, Supplementary Material

A. Per-edge and joint coverage

We make the coverage claim of §2 precise and state its assumptions, since the window W is meant as a confidence statement. Two assumptions. (A1) Under a normal approximation, each input study's reported 95% CI is a Wald interval for the effect-size parameter on that edge, with endpoints $s \pm z_{1-\alpha/2} \sigma$ at $\alpha = 0.05$. (A2) The solver's per-edge band is that interval scaled by the feasibility factor c , namely $s \pm c z_{1-\alpha/2} \sigma$.

Per-edge coverage. Under (A1)–(A2), the band of half-width $W_e = c z_{1-\alpha/2} \sigma_e$ on edge e has marginal coverage $\text{cov}_e = 2\Phi(c z_{1-\alpha/2}) - 1 = \text{erf}(c z_{1-\alpha/2}/\sqrt{2})$. At $c = 1$ this is exactly the nominal 0.95.

Joint coverage. Treating n edges as simultaneous and choosing c so that $c z_{1-\alpha/2} = z_{1-\alpha/(2n)}$ gives family-wise coverage at least $1 - \alpha$ by the Bonferroni inequality, for any dependence among edges. For the demonstration network ($n = 166$) this is $c \approx 1.85$.

Two caveats stated plainly. The normal approximation (A1) is the standard one for log-RR/OR/HR endpoints and is exact only asymptotically; it is least accurate at the smallest, lowest-base-rate cells, which is consistent with the abstention behaviour of §5. And because W is a property of the constraint polytope, these statements concern the input bands the polytope is built from: the runs in the body use $c = 1$ (exact marginally, conservative jointly), and any target joint coverage can be regenerated through the Bonferroni relation without re-solving the structure.

B. Monotonicity of the validation window

This appendix states the per-study coverage property formally and proves that the validation window is monotone under evidence addition: a consistent literature constraint can only narrow the window, a binding one strictly narrows it, and an inconsistent one inflates it. We separate two scalars that the main text folds into the single band factor c of App. A: the Bonferroni multiplier $\kappa \geq 1$ (fixed; $\kappa = 1$ gives the marginal 95%, $\kappa \approx 1.85$ gives joint 95% for our $n = 166$ studies), and the binary-searched feasibility scalar $\omega \in (0, 1]$ (the smallest scaling at which the constrained QP stays feasible, whose per-edge projection is the window W). The argument below tracks how ω moves as constraints are added at fixed κ .

Setup. For a non-root node Y with parents $X_1, \dots, X_K$, write $\mathbf{x} \in \mathbb{R}^M$ for the flattened CPT. A literature study i reports a point estimate s_i and 95% half-width CI_i for the marginal $P(Y=y \mid X_{p(i)}=v_i) = R_i^\top \mathbf{x}$, where R_i encodes the law-of-total-probability weights over the network's joint structure. The QP's per-study windowed band is $s_i - \kappa \omega \text{CI}_i \leq R_i^\top \mathbf{x} \leq s_i + \kappa \omega \text{CI}_i$. Cross-CPT subset constraints from the law of total probability take the form $S_j^\top \mathbf{x} = c_j$; CPT validity adds $\mathbf{x} \geq 0$ and the row-stochasticity equalities. The feasible polytope $\mathcal{P}(\omega, \kappa)$ is their intersection, and the feasibility band of study i is $W_i(\omega, \kappa) = \frac{1}{2}(\max_{\mathbf{x} \in \mathcal{P}} R_i^\top \mathbf{x} - \min_{\mathbf{x} \in \mathcal{P}} R_i^\top \mathbf{x})$. The solver determines $\omega^* = \min\{\omega : \mathcal{P}(\omega, \kappa) \neq \emptyset\}$ by binary search on $[0, 1]$ and reports $W_i^* = W_i(\omega^*, \kappa)$.

Theorem 1 (Per-study coverage). Assume study i 's literature 95% CI is well approximated by a normal interval at level $\alpha = 0.05$. Then for any $\kappa \geq 1$ the band $s_i \pm \kappa \omega^* \text{CI}_i$ is a confidence interval for $R_i^\top \mathbf{x}^*$ at level $1 - \alpha_\kappa$, where $\kappa z_{1-\alpha/2} = z_{1-\alpha_\kappa/2}$ and z_q is the standard-normal q -quantile. In particular $\kappa = 1$ recovers the marginal 95%, and $W_i^* \leq \kappa \omega^* \text{CI}_i$.

Proof. Under the normal approximation the half-width CI_i equals $z_{1-\alpha/2} \text{SE}_i$ for the study's standard error SE_i ; multiplying by κ widens it to $\kappa z_{1-\alpha/2} \text{SE}_i = z_{1-\alpha_\kappa/2} \text{SE}_i$, which is the stated relation. The bound $W_i^* \leq \kappa \omega^* \text{CI}_i$ holds because W_i is the half-range of $R_i^\top \mathbf{x}$ over $\mathcal{P}(\omega^*, \kappa)$, which is contained in the band; the polytope is narrower than the band whenever other constraints (LoTP subset, simplex) are active on study i 's axis. $\square$

Convergence. The binary search on ω is a one-dimensional bisection over a convex feasibility set, so ω^* is unique given the constraints; across 10 independent initialisations ω^* coincides to within solver tolerance (10^{-3} binary-search resolution, 10^{-6} in the underlying OSQP solve) on every per-cell QP.

Theorem 2 (Joint coverage). For n literature constraint rows tested at common nominal level $\alpha = 0.05$, the multiplier κ giving joint 95% coverage by Bonferroni adjustment satisfies $\kappa z_{1-\alpha/2} = z_{1-\alpha/(2n)}$. For the demonstration network ($n = 166$) this gives $\kappa \approx 1.85$.

Proof. By the union bound, $\Pr(\text{all bands cover their truth}) \geq 1 - n\alpha_\kappa$ with α_κ from Theorem 1; the bound holds under arbitrary dependence, and positive correlation among studies sharing reference cohorts (NHANES, UK Biobank) only makes coverage more conservative. Setting $1 - n\alpha_\kappa = 0.95$ gives $\alpha_\kappa = 0.05/n = 3.01 \times 10^{-4}$ for $n = 166$, so $z_{1-\alpha_\kappa/2} \approx 3.61$ and $\kappa \approx 3.61/1.96 \approx 1.85$. A Šidák adjustment is indistinguishable at this n . This is the band factor $c \approx 1.85$ of App. A; the body's runs use $\kappa = 1$ (exact marginally, conservative jointly), and any target joint coverage follows from this relation without re-solving. $\square$

Theorem 3 (Strict monotonicity of W under evidence addition). Let $\mathcal{P}_n = \mathcal{P}(\omega_n^*, \kappa)$ denote the feasible polytope after n literature constraints have been imposed at the binary-searched feasibility scalar ω_n^* , and let $W_e^{(n)}$ denote the feasibility band of edge e at stage n (as in Theorem 1). Write $\mathcal{C}_{n+1} = \{x : s_{n+1} - \kappa \omega_n^* \text{CI}_{n+1} \leq R_{n+1}^\top x \leq s_{n+1} + \kappa \omega_n^* \text{CI}_{n+1}\}$ for the slab of the new study at scaling ω_n^* .

1. (Weak monotonicity, consistent case.) If constraint $n+1$ is consistent with $\mathcal{P}_n$, i.e. $\mathcal{P}_n \cap \mathcal{C}_{n+1} \neq \emptyset$, then $\omega_{n+1}^* \leq \omega_n^*$, $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$, and consequently $W_e^{(n+1)} \leq W_e^{(n)}$ for every edge e .
2. (Strict monotonicity, binding consistent case.) If in addition $\mathcal{C}_{n+1}$ is binding on edge e (its bounding hyperplane properly separates at least one extremal point of $\mathcal{P}_n$ along the edge- e axis from the interior of $\mathcal{C}_{n+1}$), then $W_e^{(n+1)} < W_e^{(n)}$.
3. (Inflation under inconsistency.) If $\mathcal{P}_n \cap \mathcal{C}_{n+1} = \emptyset$ at scaling ω_n^* , then $\omega_{n+1}^* > \omega_n^*$ and $W_e^{(n+1)} > W_e^{(n)}$ for at least one edge e touched by the new or an active prior constraint.

Proof of part 1. Fix the feasibility scalar at ω_n^* . The feasible set at stage $n+1$ at this scalar is $\mathcal{P}_n \cap \mathcal{C}_{n+1}$, an intersection of convex sets ($\mathcal{P}_n$ by induction, $\mathcal{C}_{n+1}$ a single slab), nonempty by hypothesis, so the stage- $(n+1)$ QP is feasible at ω_n^* . By definition of ω_{n+1}^* as the minimum feasibility scalar, $\omega_{n+1}^* \leq \omega_n^*$. Then $\mathcal{P}_{n+1} = \mathcal{P}(\omega_{n+1}^*, \kappa) \subseteq \mathcal{P}_n \cap \mathcal{C}_{n+1} \subseteq \mathcal{P}_n$ (the first inclusion because a smaller ω shrinks every slab). For any edge e , since $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$ the supremum of $R_e^\top x$ over the smaller set is no larger and the infimum no smaller, hence $W_e^{(n+1)} \leq W_e^{(n)}$. $\square$

Proof of part 2. Let $x^* \in \arg \max_{x \in \mathcal{P}_n} R_e^\top x$, so $R_e^\top x^* = \sup_{x \in \mathcal{P}_n} R_e^\top x$. The binding hypothesis gives $x^* \notin \mathcal{C}_{n+1}$, hence $x^* \notin \mathcal{P}_{n+1}$. By part 1, $\mathcal{P}_{n+1} \subseteq \mathcal{P}_n$, so $\sup_{x \in \mathcal{P}_{n+1}} R_e^\top x$ is attained at some x^{**} with $R_e^\top x^{**} \leq R_e^\top x^*$. Because the separating hyperplane of $\mathcal{C}_{n+1}$ strictly separates x^* from $\mathcal{P}_{n+1}$ and (by the binding hypothesis) the separation direction has nonzero component on the edge- e axis, the inequality is strict, $\sup_{x \in \mathcal{P}_{n+1}} R_e^\top x < \sup_{x \in \mathcal{P}_n} R_e^\top x$. The same argument applies to the infimum if the new constraint also binds there; in either case the strict inequality propagates to the projection width, $W_e^{(n+1)} < W_e^{(n)}$. $\square$

Proof of part 3. If $\mathcal{P}_n \cap \mathcal{C}_{n+1} = \emptyset$ at ω_n^* , both sets are closed convex and a separating hyperplane exists. Each slab half-width is linear in ω , so $\mathcal{C}_{n+1}(\omega)$ and every prior slab in $\mathcal{P}_n(\omega)$ expand with ω , and joint feasibility $\mathcal{P}_n(\omega) \cap \mathcal{C}_{n+1}(\omega) \neq \emptyset$ becomes more permissive as ω grows. By continuity of the boundaries in ω there is a unique minimum $\omega_{n+1}^* > \omega_n^*$ restoring feasibility, at which at least one slab (the new constraint, or a prior constraint active at the old ω_n^*) has its width strictly inflated, and the corresponding edge satisfies $W_e^{(n+1)} > W_e^{(n)}$. $\square$

Remark (objective-invariance). Theorem 3 concerns only the constraint polytope. The QP's least-squares objective and monotonicity penalty (§2) select a single point inside $\mathcal{P}_n$ but do not enter the definition of W_e , so W_e is invariant to the objective weight λ , as stated in §3 and consistent with Theorem 1.

Remark (median). Part 1 gives pointwise weak monotonicity per edge and part 2 strict monotonicity wherever the new constraint binds; together they imply $\tilde{W}^{(n+1)} \leq \tilde{W}^{(n)}$ for the median, with strict inequality whenever the new constraint binds on at least half the active edges. This is the formal underpinning of the median-window reduction reported in §4 (Table 1): each consistent audit edge binds a substantial fraction of the median edges' projections.

C. Determination of untested parent combinations: the LoTP activity threshold

Section 3, Property (4) asserts that the law-of-total-probability subset constraints determine untested parent combinations to within W , but only when the parent co-occurrence ρ is large enough to make the constraints tight. We make this precise.

Setup. Let Y be a binary node with K binary parents $X_1 \dots X_K$ and let $\mathbf{x} \in \mathbb{R}^{2^K}$ be its flattened CPT. Each marginal study i gives a linear equality $r_i^\top \mathbf{x} = s_i \pm \sigma_i$, where σ_i is the half-width of the literature 95% CI in CPT-cell units. Each subset constraint is a linear identity holding the law of total probability over a group of cells parametrised by the reference-survey co-occurrence ρ . Concretely, for two parents the law of total probability reads

$$P(Y \mid X_i=v_i) = P(Y \mid X_i=v_i, X_j=v_j) p + P(Y \mid X_i=v_i, X_j \neq v_j) (1-p),$$

with $p = P(X_j=v_j \mid X_i=v_i)$. Writing $m = P(Y \mid X_i=v_i)$ for the literature marginal and ranging the cofactor cell over its simplex bound $[0, 1]$ pins the unknown joint cell to the interval $[\max(0, \frac{m-1+p}{p}), \min(1, \frac{m}{p})]$, of pre-clamp width $(1-p)/p$. Let σ denote the binding noise scale (the smallest σ_i relevant to the subset constraint).

Theorem (constraint-bite, informal). When $|\rho|$ exceeds the threshold

$$|\rho^*| = \frac{\sigma}{1+\sigma}, \quad (2)$$

the subset constraint pins each untested cell to within a half-width of $W \leq \frac{1-\rho}{1+\rho} \sigma$. Below threshold the subset constraint is non-binding and W relaxes back to the marginal-only width σ .

Sketch. The subset constraint is linear in ρ ; for two parents the off-diagonal cells are determined by a 2×2 block whose condition number is $(1+\rho)/(1-\rho)$. The constraint binds when this conditioning exceeds the marginal-noise budget, which after rearrangement is precisely (2). At $\rho=\rho^*$ the subset row enters the active set of the QP; for $\rho > \rho^*$ the active-set boundary tightens as $(1-\rho)/(1+\rho)$, giving a hard half-width on the combination cell. The multi-parent extension takes the relevant ρ as the median over parent pairs.

Rigorous threshold. The heuristic (2) is the envelope of a rigorous threshold $\varphi(\sigma, K, p)$ that depends on the parent count K , the parent marginal p , and σ (the input-CI half-width as a fraction of the cell value, the QP's effective noise scale). We prove it under the equicorrelated Bahadur-1 multivariate Bernoulli model for the parent joint, with one literature row per parent; heterogeneous pairwise correlations are handled as a perturbation (Corollary 1), and higher-order parent-joint moments contribute a controlled error term discussed in the scope statement.

The Bahadur-1 ansatz. Under the equicorrelated Bahadur-1 model every parent has the same marginal p , every parent pair the same correlation ρ , and all higher-order central moments vanish. The joint takes the form $\pi_{\mathbf{v}} = \nu_{\mathbf{v}}^{\text{ind}} (1 + \rho \sum_{i < j} \psi_{ij}(\mathbf{v}))$, where $\nu_{\mathbf{v}}^{\text{ind}} = p^{|\mathbf{v}|} (1-p)^{K-|\mathbf{v}|}$ is the independence joint, $z_k(v_k) = (v_k - p) / \sqrt{p(1-p)}$ is the standardized indicator, and $\psi_T(\mathbf{v}) = \prod_{k \in T} z_k(v_k)$ is the level- $|T|$ Bahadur basis function. The Bahadur basis is orthonormal under ν^{ind} ; because Y 's constraints lie in the level- ≤ 2 subspace, the constraint-matrix analysis localises to a $(1 + K + \binom{K}{2})$ -dimensional subspace.

Network ρ vs. reference-survey ρ . Let $\hat{\pi}$ be the parent joint identified by the QP from joint coverage in the reference population; the network pairwise correlation ρ_{ij}^{net} is its Bahadur level-2 coefficient. When joint coverage is sparse on a parent set, $\hat{\pi}$ collapses onto the support of available joint observations and ρ_{ij}^{net} may differ from the marginal NHANES pairwise correlation $\rho_{ij}^{\text{NHANES}}$; the threshold below is stated in ρ^{net} , and for densely covered parent sets $\rho^{\text{net}} \approx \rho^{\text{NHANES}}$.

Theorem 4 (LoTP activity threshold, $K \geq 3$). Under the equicorrelated Bahadur-1 model with $K \geq 3$ binary parents (common marginal p , common pairwise correlation ρ^{net}), K literature rows (one per parent), per-output NHANES marginal, and at least one LoTP cross-CPT subset row S whose level-2 Bahadur projection has non-zero component orthogonal to $\text{span}(A_{\text{lit}}^{(2)})$, the LoTP constraint is active on at least one literature axis (i.e.

$W_i^* < \kappa\omega^* \text{CI}_i$ for at least one i) iff $\rho^{\text{net}} > \varphi(\sigma, K, p)$, where $\varphi(\sigma, K, p)$ is the unique root in $(0, 1)$ of

$$\rho \cdot \eta(K, p) (1 - \theta_{\text{LoTP}}) = \kappa\omega s_{\text{max}} \cdot \min((1 - \rho) B_-(p), \rho \tau_2^+(K, p)). \quad (3)$$

The threshold admits a two-regime characterisation,

$$\varphi(\sigma, K, p) = \begin{cases} \frac{\sigma}{\sigma + \mu(K, p)}, & \sigma \leq \sigma^*(K, p) \quad (\text{level-1 binding}) \\ \frac{1}{1 + \sigma \mu'(K, p)}, & \sigma > \sigma^*(K, p) \quad (\text{level-2 binding}) \end{cases}$$

with crossover $\sigma^*(K, p) = B_-(p) / [\kappa\omega s_{\text{max}} \tau_2^+(K, p)] \cdot \mu / (\mu + \mu')$. Here $B_-(p) = \sqrt{\min(p, 1-p) / \max(p, 1-p)}$ is the level-1 spectral floor, $\tau_2^+(K, p)$ is the smallest singular value of the level-2 sub-block of the constraint matrix after LoTP rows are added, $\eta(K, p)$ is the level-2 norm of the literature row R_i , and $\theta_{\text{LoTP}} \in [0, 1]$ is the fraction of R_i 's level-2 signal absorbed by the LoTP row. Direct numerical computation of $\sigma_{\min}$ on the network's A_{lit} matrices gives $\mu(K, p), \mu'(K, p) \in [0.5, 2.5]$ across the operational range $K \in \{3, 4, 5, 6\}$, $p \in [0.1, 0.5]$.

Connection to the empirical envelope. The body operates at small literature-CI noise ($\sigma \approx 0.10$ – 0.20), the level-1-binding regime, where $\varphi = \sigma / (\sigma + \mu)$ with $\mu \in [0.5, 2.5]$; the heuristic $\sigma / (1 + \sigma)$ of (2) is the $\mu = 1$ case, correct as a multiplicative-factor-of-2 envelope of φ across the operational range, which licenses its use in the body. (The level-2 form $1 / (1 + \sigma \mu')$ governs only the large- σ regime $\sigma > \sigma^*$, outside the paper's operating point.)

Proof sketch. Step 1 (constraint matrix in the Bahadur basis). The literature row R_i clamping parent X_p at value a has Bahadur coefficients $\alpha_\emptyset = 1$, $\alpha_{\{p\}} = \beta_a$, $\alpha_{\{j\}} = \rho \beta_a$ for $j \neq p$, $\alpha_{\{i,j\}} = \rho$ for $i, j \neq p$, $\alpha_{\{p,j\}} = \rho \beta_a^2$ for $j \neq p$, and $\alpha_T = 0$ for $|T| \geq 3$, with $\beta_a = (a - p) / \sqrt{p(1-p)}$. The per-output marginal row m_Y has $\alpha_\emptyset = 1$, $\alpha_{\{i,j\}} = \rho$, else zero; the LoTP row S also has level cap ≤ 2 . Step 2 (level-stratified $\sigma_{\min}$). The level-1 sub-block of A_{lit} has structure $\text{diag}(\beta_{a_1}, \dots, \beta_{a_K}) ((1 - \rho) I_K + \rho \mathbf{1}\mathbf{1}^\top)$, whose smallest singular value is $(1 - \rho) B_-(p)$. The level-2 sub-block has every entry proportional to ρ , giving $\sigma_{\min}(A_{\text{lit}}^{(2)}) = \rho \tau_2(K, p)$, so $\sigma_{\min}(A_{\text{lit}}) = \min((1 - \rho) B_-(p), \rho \tau_2(K, p))$. Step 3 (effect of the LoTP row). Adding S produces A_{full} with $\sigma_{\min}(A_{\text{full}}^{(2)}) = \rho \tau_2^+(K, p)$, $\tau_2^+ > \tau_2$ (since S 's level-2 projection has component orthogonal to $\text{span}(A_{\text{lit}}^{(2)})$). The polytope-projection bound (Vavasis and Ye, 1996; Dadush et al., 2023) gives $W_i \asymp \|P_{\ker A_{\text{full}}} R_i\| / \sigma_{\min}(A_{\text{full}})$. With $\|P_{\ker A_{\text{full}}} R_i\| = \rho \eta(K, p) (1 - \theta_{\text{LoTP}})$, the activity criterion $W_i \leq \kappa\omega \sigma s_i$ rearranges to (3), whose unique positive root is $\varphi(\sigma, K, p)$; the two-regime form follows from which term in $\min((1 - \rho) B_-, \rho \tau_2^+)$ binds, with crossover at $\sigma^*(K, p)$. $\square$

Corollary 1 (Heterogeneous pairwise correlations). Drop equicorrelation: let ρ_{ij}^{net} have median ρ_{med} and spread $\Delta_\rho := \max_{ij,kl} |\rho_{ij}^{\text{net}} - \rho_{kl}^{\text{net}}|$. Provided the maximum sub-determinant of the parent-correlation matrix is bounded away from zero, the threshold $\rho_{\text{med}} > \varphi(\sigma, K, p)$ continues to govern activity, with $|\text{exact threshold} - \varphi(\sigma, K, p)| \leq C(K, p) \Delta_\rho$, where $C(K, p) = O(\sqrt{K} / \tau_2(K, p))$ is the matrix-perturbation Lipschitz constant of Tropp (2015, Cor. 6.1.6). This licenses using the median pairwise correlation as the operative quantity in §3 despite heterogeneous reference-survey correlation matrices.

Theorem 5 ($K = 2$ base case). Under the Bahadur-1 model with $K = 2$ binary parents, two literature rows, per-output NHANES marginal, and one LoTP cross-CPT subset row S with non-degenerate level-2 component, the LoTP constraint is active iff $\rho^{\text{net}} > \sigma / (\sigma + \mu_2(p))$, where $\mu_2(p) \in [0.6, 1.4]$ for $p \in [0.1, 0.5]$. In our dataset $K = 2$ targets typically have one literature row per parent, so the 4-cell CPT is small-cell-determined by literature plus per-parent and per-output marginals plus simplex; the theorem is informative when fewer than two literature rows are available, and is included for derivational continuity with Theorem 4.

Scope and failure modes. Theorem 4 is rigorous under the equicorrelated Bahadur-1 ansatz. Five scope items. (i) Higher-order joint structure: the ansatz assumes parent-joint moments of order ≥ 3 vanish; real reference-survey joints have non-zero higher moments, introducing relative error $\sum_{k \geq 3} \rho_k^2 / \rho_2^2$ (ρ_k the L^2 -norm of the level- k block), empirically small on the parent sets we care about ($K \leq 5$). (ii) Heterogeneous marginals: equicorrelation assumes common p ; heterogeneous p_k destroys the closed-form spectrum, and Corollary 1 covers heterogeneous correlations, not marginals. (iii) Missing literature on some parents: when the literature-row count is below K , the level-1 sub-block is rank-deficient and the level-2-binding regime governs. (iv) LoTP rows with degenerate

level-2 component: a LoTP row from a downstream child sharing all K parents with Y can have its level-2 projection vanish, making LoTP structurally inert; the theorem applies to LoTP rows conditioning on a strict parent subset. (v) The constant θ_{LoTP} : the level-2 absorption fraction depends on the LoTP geometry, is generically $\Theta(1)$, and is treated as known by direct $\sigma_{\min}$ computation, absorbed into μ, μ' .

Related work on the threshold. The activity-threshold result sits at the intersection of three literatures. Credal networks and imprecise-probability propagation: Cozman (2000) characterises the strong-extension fixed point but gives no quantitative activity threshold; the CREMA library (Huber et al., 2020) implements constraint propagation as an algorithmic, not a constraint-matrix, property; the imprecise Dirichlet model (Walley, 1996) gives concentration intervals under i.i.d. multinomial data, not correlated parents. LP polytope geometry: the Vavasis–Ye $\bar{\chi}_A$ -condition (Vavasis and Ye, 1996; Dadush et al., 2023) underlies Step 3’s polytope-projection bound; the Restricted Isometry Property literature (Candès–Tao 2005) preserves geometry under correlated random matrices but assumes isotropy/sub-Gaussianity, unlike our deterministic Bahadur structure. Constrained Bayesian-network learning: parameter-equality constraints (Niculescu et al., 2006), monotonicity constraints (Feelders–van der Gaag 2006), and qualitative sign-propagation (Wellman, 1990) address hard constraints, not the soft CI-band activity our threshold characterises.

Practical reading. For typical literature CIs ($\sigma \approx 0.10$ – 0.20) the bite threshold is $\rho^* \approx 0.13$ – 0.17 . Parent pairs whose reference-survey co-occurrence reaches $|\rho| \geq 0.20$ are in the firm-bite regime: their combination CPT cells are determined to within W without any joint study. Pairs whose $|\rho| < 0.15$ are reported as wide-window W and surfaced on the widest- W list (Property 2): the system reports that it cannot extrapolate without joint data. Of the 336 parent-pair edges for which NHANES has ≥ 30 co-occurrence observations, 30% (102/336) reach $|\rho| \geq 0.15$ and 25% (84/336) reach $|\rho| \geq 0.20$; these resolve their combination cells to $W \leq 0.05$. The remainder cluster on the widest- W list and prompt either an additional connecting study or a structural change (a latent variable bridging the correlated co-parents). Median NHANES $|\rho|$ across all observable parent pairs is 0.044, so the bite regime is the upper tail of the distribution; but it is precisely those upper-tail edges that drive the window reduction of §4.

Connection to identifiability. The classical question of when marginal effects suffice to determine joint conditionals here receives a per-edge answer: the bite threshold (2) is the per-edge identifiability threshold, W is the per-edge non-identifiability radius, and the widest- W list is the sorted ranking of where in the DAG more evidence is needed. Identifiability is not a binary attribute of the model; it is a feasibility radius reported per edge and updated each cycle.

Empirical verification. Joining per-edge W with NHANES-measured $|\rho|$ on the improved build (194 edges) gives a step at the predicted threshold: median $W = 0.25$ for $|\rho| < 0.15$ ($n=150$, no bite); $W = 0.13$ for $[0.15, 0.20)$ ($n=8$, partial); $W = 0.001$ for $[0.20, 0.30)$ in QP-solved edges ($n=19$, full bite). Loose edges in the bite regime are any_of aggregators or equivalent_to wrappers, correctly outside the theorem’s domain.

D. Causal direction and confounding scope

The construction is causally directed. The DAG’s edge directions are set by LLM reading of the agreed-upon causal mechanisms in the biomedical literature (review articles, mechanistic textbooks, and clinical-reasoning sources describing which exposures drive which outcomes through which intermediates). The LLM does not pick directions from association strength on the survey; it picks them from the literature’s mechanistic consensus, the same channel a domain expert would use. Where that consensus is contested (depression $\leftrightarrow$ coronary artery disease), it takes the direction the majority of sources emphasise, and the residual disagreement surfaces later as a wide validation window or a direction-check failure. Direction-setting, edge extraction, and the structural placement of mediating nodes are the three things the LLM supplies that statistical analysis of the survey alone cannot. The result is a causal DAG that is encoded, not discovered: we run no causal-discovery algorithm over the data.

On confounding the construction is deliberately conservative. The network reported here fits the literature’s observational RRs as published, without per-edge confounder adjustment. Two structural tools bear on confound-

ing. dependency_distal wrappers (App. I) install a literature-backed correlation between two sibling parents the network would otherwise treat as independent (a shared-child v-structure), repairing false independence and countering aggregator washout while holding the disease’s parent count fixed; this is a dependence-repair device, not a mediation decomposition, and true mediators are modelled separately as explicit $A \rightarrow M \rightarrow D$ chains. Separately, the software provides an optional, off-by-default deconfounding mode that, on the oriented DAG, converts a parent’s total RR into a direct effect via the exact log-odds identity $\log \text{OR}_{\text{direct}} = \log \text{OR}_{\text{total}} - \log \text{OR}_{\text{indirect}}$; it is disabled in this build because subtracting modelled indirect effects from already-observational RRs risks double-adjustment.

What this licenses: the reported $P(Y | X = x)$ are literature-coherent observational conditionals organised by mechanistic causal direction, not identified interventional effects $P(Y | \text{do}(X = x))$ in the sense of Pearl (1988). Meta-analytic adjustment sets across our source studies are heterogeneous and not tracked per row; the survey is cross-sectional and cannot separate forward from reverse causation on every edge; and where the mechanistic literature is wrong about a direction, the LLM inherits the error and the window cannot detect it. For deployment requiring interventional semantics, the additional no-unmeasured-confounding assumption is the deployer’s to make and disclose. For this paper’s claims (literature-coherent risk estimation, abstention where literature plus structure cannot commit, and research-direction signalling via widest- W edges), the observational conditionals are the right object, and the window reports the residual uncertainty per edge.

E. Effect-size harmonisation

The system accepts four effect-size types and normalises each to the marginal-RR form of Theorem 1 before entering the QP.

OR $\rightarrow$ RR. Standard conversion $\text{RR} = \text{OR} / (1 - p_0 + p_0 \text{OR})$ using the target output’s NHANES marginal p_0 . For common outcomes ($p_0 \gtrsim 0.10$) this conversion can systematically overstate the effect, a known contributor to descriptive-tier residual error (App. M).

HR $\rightarrow$ RR. Hazard ratios are treated as time-stationary RRs without explicit conversion to a fixed-horizon cumulative incidence. HRs from studies with different follow-up windows are not directly comparable; the QP absorbs this incomparability into the residual rather than detecting it.

SMD $\rightarrow$ RR. Standardised mean differences are mapped to RR via the standard normal-link approximation with a guard against extreme values ($|\text{SMD}| > 1$ flagged as a single-trial high-variance warning, blanked when the source is a single small RCT).

These harmonisations are sources of systematic bias the QP cannot detect from the constraints alone; they enter as bias rather than variance, and the feasibility band W does not bound them. The distinction matters for interpretation: W reports residual disagreement among the literature constraints as they reach the QP, not residual disagreement among the underlying epidemiological designs that produced them.

F. LLM-extraction validation

Because the spreadsheet rows are produced by an LLM reading published studies, extraction error is a primary risk, and three controls bound it. (1) Blind re-extraction. A second LLM agent, given a differently phrased prompt, independently re-extracts the effect size, CI, population, and PMID for each edge; rows where the two agents disagree on the effect size beyond a tolerance, or name different source papers, are held for human adjudication rather than admitted. (2) Citation verification. Every cited PMID is resolved against PubMed and its title and abstract are checked for topical match to the encoded relationship; rows whose PMID does not resolve, or resolves to an unrelated paper, are dropped. An audit of the LLM-extracted corpus under this procedure found and corrected a minority of rows carrying mis-resolved citations or out-of-tolerance effect sizes. (3) Two solver-side catches that need no external ground truth. Each edge’s encoded direction is checked against the network’s query-RR, so a sign flip is surfaced (§4); and the law-of-total-probability subset constraints reject any set of cells that cannot be made jointly coherent. A mis-transcribed effect size therefore appears either as a direction-check failure or as a widened validation window, so the extraction step needs only to be auditable, and

W is the audit channel.

Audit results. We hand-validated a random sample of 30 study rows drawn uniformly without replacement (`random.seed = 7`) along five dimensions: paper existence, title match, journal match, year match, and semantic correspondence between the CSV’s input variable and the paper’s exposure. Of the 30, 29 resolved to real, verifiable PubMed-indexed papers; one had no citation and was removed pending re-extraction. Five rows carried non-substantive metadata noise (journal-field errors, year off by one or three, paraphrased rather than verbatim titles, one design-code mismatch), corrected without affecting the QP, which consumes only the statistic, its CI, and the parent-child structure. One row applied a smokeless-tobacco effect estimate to an input variable labelled `ever_vaped` (pharmacologically distinct exposures sharing the surface form “tobacco-adjacent”), and was corrected. A second random sample (`random.seed = 11`, no overlap) gave 30/30 bibliographic accuracy and 28/30 semantic match. Combined bibliographic accuracy across both samples is 59/60 = 98.3% (95% Clopper–Pearson [91.1%, 99.96%]); semantic match was 28/30 on the second sample, with the first sample reported below under the more conservative independent re-extraction (we do not pool the two semantic counts, which use different protocols).

Self-critical finding from independent re-extraction. An independent re-extraction of the same $n=30$ first sample by a separate LLM identified four substantive errors the single-pass audit missed: one wrong-paper attachment, two U-shape papers where the LLM took one tail’s HR for the opposite-tail row, and one paraphrased-vs-mismatched title at PMID 27744349. This puts the substantive-fidelity rate at 26/30 on the first sample under independent re-extraction, below the 29/30 of the single pass; we do not claim the remaining unchecked rows are error-free at the higher rate. The window W and the per-row direction test are the operational catches for errors of this kind: the architectural assumption is that the LLM needs to be auditable rather than perfect, with W as the audit channel.

G. Behaviour under sparse or contradictory evidence

Two regimes deserve their own treatment. Contradiction. When two studies on the same edge report incompatible effect sizes, or an edge’s literature contradicts an adjacent edge’s literature or the population data, the feasible polytope cannot satisfy every band at the literature-CI width, so the feasibility factor c must widen and the edge’s validation window exceeds its own literature CI. The system does not average the conflict away; it reports it, and the widest-window list (§4) ranks exactly these tensions for resolution. Sparsity. When an edge has few or no constraining studies and few co-occurring parents, the polytope is wide and W is correspondingly large; at very low base rate the polytope becomes nearly symmetric along the direction axis and the solver abstains rather than commit to a sign (§5: the pancreatic-cancer chain at base rate $\approx 2 \times 10^{-4}$, query RR 0.055 against literature 1.3–1.7). Both behaviours are by design: contradiction widens W , sparsity widens W or triggers abstention, and in neither case does the system present a confident answer where it lacks the evidence to support one.

H. Wiring rules and silent-zero propagation

The construction guidance of §6 reduces to wiring rules that keep the polytope informative as edges are added, each preventing a specific saturation or inversion failure mode (App. I); the rules are enforced by structural data-checks run before every build.

Four wiring rules (logical gates). (R1) Trigger each leaf on its least-prevalent state, with the RR direction set accordingly: rare-side triggers carry information, same-side triggers saturate the gate. (R2) Limit K per gate. An `any_of` gate collapses K inputs to one yes/no, losing more information at higher K ; target $K = 2\text{--}3$, hard ceiling $K = 7$. (R3) Do not group correlated leaves into one gate. Leaves with $|\rho| \geq 0.30$ saturate the OR-mask and lose the cross-leaf evidence flow the QP would otherwise use; cluster them under a common-cause latent or a multi-rung wrapper ladder (chain-depth note below). (R4) Do not mix opposite-direction RRs in one gate. Risk inputs ($RR > 1$) and protective inputs ($RR < 1$) routed to the same `any_of` cancel via the QP fit, producing direction-inverted residual signal; split them into separate sub-gates.

Five within-aggregator sibling-discipline rules. Within one aggregator or wrapper, all sibling rows must satisfy: (S1) same direction on the disease (mixed signs cancel); (S2) near-magnitude RRs (strong siblings dominate the QP fit; empirical envelope $\max / \min \leq 3$); (S3) low-prevalence triggers (states with marginal $P > 0.30$ saturate the OR-mask); (S4) effect traced empirically (set evidence on the leaf, query up the chain, confirm the direction preserves to the disease); (S5) semantic coherence (siblings represent the same kind of thing the gate names; a physical-activity sibling does not belong inside a healthy-diet aggregator even though both predict diabetes, since semantic confusion produces phantom signal flows that double-count when the node also feeds its proper gate).

Structural data-checks. The grammar is enforced by automated checks (`scripts/data_checks.py`) before every build. Wrapper value-name alignment verifies that every `dependency_distal` wrapper’s exposed-state columns name states that exist as outvars on the wrapper’s primary-chain parent (a misalignment causes the silent zero-propagation below); consumer state-name reachability verifies that every consumer row’s input-value reference matches one of the wrapper’s exposed states. Other checks enforce $K \leq 7$, no self-reference, no orphan inputs or outputs, no row missing input values, and no duplicate (output, input, input-value) tuples. An LLM that emits a violating row sees the failure message and revises before the QP solver is invoked.

Silent zero-propagation, and the trace test (S4) that catches it. One class of LLM-authoring failure is silent at the marginal level but breaks end-to-end propagation. When a `dependency_distal` wrapper’s exposed-state names do not match the outvar names of its primary-chain parent, every consumer row referencing the wrapper hits a state-name lookup miss at inference time (silently, with no error raised), and the consumer’s evidence assignment is dropped. The wrapper’s own marginal still tracks its inputs and the disease’s marginal still sits near baseline, so standard W and direction tests do not flag it: W stays small because with no propagation the literature constraint is trivially satisfied at any feasible point, and the direction test reads NEUTRAL because $P(Y \mid \text{parent}=v)$ and $P(Y)$ are both the disease’s marginal regardless of the parent. The leaf-to-disease trace test (S4) catches it: set leaf-on evidence at the bottom of the chain, query each node upward, and confirm the disease’s marginal moves in the literature-implied direction; the lookup miss surfaces as a wrapper marginal that moves with its parent combined with no movement in the disease. The two structural checks above reject the value-misalignment pattern at authoring time. The general pattern is a row that violates a sub-symbolic invariant (state-name reachability) while passing every symbolic syntax or schema rule; the construction manual is the place to encode each such invariant as a checkable property.

Chain depth trades magnitude for K . Empirically each wrapper layer between a leaf and the disease attenuates the leaf’s queried effect by roughly half, so depth trades magnitude for the K -reduction it buys; a leaf that must retain its magnitude is promoted to a direct disease parent.

I. The BayesExpert construction vocabulary

The Type column on each row declares either a node type (a node’s role and CPT) or a link type (a definitional or routing relationship between two nodes). We summarise the vocabulary; the construction manual released with the code gives the full grammar.

Node types (declared on a definition row).

- NHANES leaf: a directly-observed survey variable. A `naive_0` leaf has its CPT computed from NHANES joint counts against a parent, subject to guards against degenerate or definitionally-circular joints.
- dependency: a node whose CPT $P(Y \mid X_1, \dots, X_K)$ the QP solves from the study RRs on its parent edges (the default study-backed node), with variants for NHANES-coded and prior-only parents.
- any_of / all_of: logical aggregators whose CPT is a deterministic OR or AND truth table over their parents, collapsing many correlated leaves into a single binary node instead of K separate parents.
- `dependency_distal`: the wrapper node described below.

Link types (a relation between two nodes).

- `is_a` / subsumes: definitional containment, $A \subset B$ encoded by sensitivity = $P(A)/P(B)$, specificity = 1 (so

$P(B | A) = 1$), and its inverse.

- `equivalent_to / equivalent_distal`: identity, sensitivity = specificity = 1 (so $P(A) = P(B)$); the channel by which an NHANES measurement reaches a canonical node, with `equivalent_distal` the identity link used inside a distal wrapper.
- `distal`: a routing link whose aggregator-to-disease RR is not a study but is computed by aggregating the aggregator’s leaf RRs weighted by NHANES co-occurrence (a Monte-Carlo over the leaf CIs gives the disease-edge CI). A leaf row tagged with a disease name, e.g. `output=chemical_pc_risk`, `input=nickel_exposure`, `Type=pancreatic_cancer`, `RR=1.7`, routes that leaf’s RR (1.7) to the named disease through such a chain.

The `dependency_distal` wrapper. The distinctive construct is a three-row block that adds a connecting-study correlation between an aggregator’s output and a correlated sibling node without raising the parent count K at the downstream disease. The block is: (i) an `equivalent_distal` identity row routing the original aggregator into the wrapper unchanged (sensitivity = specificity = 1); (ii) the `dependency_distal` definition row declaring the wrapper node and its states; (iii) a regular literature-RR row whose input is the correlated sibling, pinning a marginal of the wrapper’s CPT to a literature-anchored value that propagates into the disease’s marginal through the law of total probability. The wrapper treats two regimes the plain construction handles poorly: under-determined interaction cells at multi-parent targets whose parent-pair correlation is below the LoTP-activity threshold, and signal attenuation in long aggregator chains (each wrapper layer roughly halves the queried effect, App. H), where anchoring the wrapper’s CPT against a sibling restores literature-shape behaviour however attenuated the upstream signal. The disease still sees a single distal-typed parent, so the cross-edge correlation is added without the 2^K cost of an extra direct parent.

J. Per-respondent ROC AUC and discrimination metrics

The metrics here are computed on the post-submission improved network (573 nodes), which extends the 427-node build of Table 1 with corrected and additional literature and resolves the degenerate conditional tables present in earlier builds. We report them as supplementary post-submission evidence, not as part of the reviewed results. The direction-accuracy and window figures below use a corrected per-edge reference-category metric and are not directly comparable to the paper-era figures of Table 1; the residual direction inversions concentrate in the low-base-rate cancer chains that §5 characterises as abstentions.

Per-target discrimination. On held-out NHANES respondents, with each target’s own NHANES code and its definitional-surrogate biomarkers excluded from the evidence set (so the model cannot read the answer off a defining measurement, App. L), per-respondent ROC AUC averages 0.732 across the 15 constructed disease targets of §4 (range 0.558–0.940). We exclude the directly-observed fall-history outcome, which has no constructed structure to discriminate and merely sits at its base rate. Several targets reach or exceed the level of cohort-fitted clinical risk calculators (≈ 0.71 for Framingham 10-year CVD and pooled ASCVD): coronary artery disease 0.94, depression 0.83, heart attack 0.82, insomnia 0.81, sleep apnea 0.80, diabetes 0.79, and stroke 0.76; the remainder fall below calculator level, down to asthma at 0.56. Per-target AUC tracks whether the network’s literature supplied informative risk-factor parents for the target. Table 2 gives the full breakdown.

Complete measurement panel. On the same build: direction accuracy 91.7% (the query-RR lands within a factor of two of the literature RR on 85.3% of edges); median validation window 0.001 (mean 0.086); calibration ECE 0.023 and MCE 0.40; Brier score 0.092; prediction sharpness (variance) 0.028; and mean $|\rho\text{-gap}|$ 0.098. Sex-stratified AUC is 0.735 (male) versus 0.696 (female), reflecting the network’s women’s-health-weighted literature. Of 356 internal nodes, 77 remain under-determined by the current literature; their predictions revert to the solver’s prior, where the window reports the residual uncertainty.

Signal across chain length. A stricter measure asks whether a study’s directional signal is retained along the full path from leaf to disease, counting a flattened (no-movement) edge as a failure rather than as not-wrong. Resolved by the number of hops from evidence to target, it declines with depth: 195/220 at one hop, 110/122 at two, 37/44 at three, 17/30 at four, and 4/10 at five (363/426, 85%). It sits below the panel’s 91.7% direction accuracy because that figure does not penalise no-movement edges and also covers the ten edges with no leaf-to-disease path; the flattened fraction rises with depth as the law-of-total-probability signal weakens across more

Target	AUC	Prevalence
Coronary artery disease	0.94	0.057
Depression	0.83	0.063
Heart attack	0.82	0.043
Insomnia	0.81	0.16
Sleep apnea	0.80	0.037
Diabetes	0.79	0.07
Stroke	0.76	0.03
COPD	0.73	0.053
Cognitive impairment	0.72	0.16
Hypertension	0.72	0.28
Sleep disorder	0.69	0.09
Cancer	0.64	0.097
Malnutrition	0.63	0.05
Osteoporosis	0.56	0.07
Asthma	0.56	0.50
Mean	0.732	

Table 2. Per-respondent ROC AUC by target on the improved network, with definitional-surrogate biomarkers excluded per App. L. Scored with `roc_auc_score` on held-out NHANES respondents ($n=300$ each). AUC tracks the informativeness of each target’s literature-supplied parents, not a ranking of the diseases.

intervening nodes.

Calibration against NHANES marginals. Mean absolute calibration error against the NHANES marginals is near-zero across the 289 variables of this improved network whose CPTs are not fit from NHANES (the leakage-free subset), an integrity check on marginal recovery distinct from the reliability ECE reported above. This count is a different criterion (leakage-free, not QP-solved) measured on a different (573-node) network than the 287 solved-CPT nodes over which the window of §4 is reported on the 427-node paper-era build, so the two are not expected to coincide.

Relation to an unconstrained baseline. The natural ablation is the same network with the subset constraints removed, and Table 1 is that comparison in miniature: with the law-of-total-probability constraints non-binding the windows are wide and uninformative, and turning them on together with consistent edges is what drives the 87% window reduction. A full unconstrained-UQ or MCMC baseline is left to future work; the credal-network framing of §3 positions the method against elicited-bound approaches rather than against a sampler.

K. UK Biobank external validation

External validation against an independent cohort. We compare the network’s query-RRs, read live from the canonical build, against published UK Biobank cohort effect sizes on 50 matched exposure-outcome pairs not used in construction. 43/50 (86%) agree in direction with the UK Biobank estimate, and 42/50 (84%) fall within 50% of it. This is a held-out comparison against a cohort distinct from the construction literature and the NHANES reference set.

L. Per-target diagnostic-biomarker exclusion

The per-respondent AUC of App. J is meaningful only if the model cannot read a target’s answer off a measurement that defines that target. We therefore exclude, per target T , every evidence node X that is a definitional surrogate of T , by a strict rule: X is dropped from the evidence for T iff it is (a) a deterministic aggregator ancestor of T ; (b) a same-NHANES-code alias of T ; (c) a diagnostic biomarker that defines the disease threshold, e.g. HbA1c, fasting glucose, and HOMA-IR for diabetes, the last excluded because it is a deterministic function of the already-excluded fasting glucose and insulin; (d) the NHANES screening item that is the disease question; or (e) for an umbrella diagnosis, a subtype reachable through the umbrella’s membership channel (the specific cancers under an all-cancer target; emphysema and chronic bronchitis under COPD). The rule deliberately does

not exclude risk factors, sibling diseases, downstream mortality, or merely-correlated clinical observations: these are legitimate evidence. Symptom items with diverse non-disease causes (individual depression-screen items, single-night sleep complaints) are likewise retained: predictive but not definitional. This separation is what distinguishes a discrimination estimate from one inflated by a biomarker that is the diagnosis under another name; without it the apparent AUC on several targets rises sharply but reports only definitional equivalence, not prediction.

M. Confidence tiers induced by the window

The validation window induces a natural confidence tiering of the network’s edges, and we find the tier predicts external generalisation. An edge (target Y) is tight-committed when its CPT is effectively determined: either small enough that literature, population marginals, and the simplex bounds supply as many equations as unknowns, or with a parent pair correlated strongly enough that the law of total probability is active ($|\rho| > \sigma/(1 + \sigma)$). It is descriptive when direction and rough magnitude are pinned but the window is wider, and abstain when the polytope is symmetric along the direction axis (§5). The external UK Biobank check (App. K) is consistent with this reading at the aggregate level (43/50 direction-concordant); at $n=50$ the sample is too small to resolve a separate per-tier generalisation rate, which we leave to a larger external cohort. The practical use is triage: a deployment trusts tight-committed edges, treats descriptive edges as directional only, and routes abstain edges to the window report rather than a point answer.

N. Crowdsourced construction and continual improvement

The construction is designed to ingest literature continuously rather than as a one-time corpus. New study rows are proposed by the LLM extractor (App. F) and may also be contributed by domain experts; each proposed row is admitted only after the two-agent quorum and PubMed verification of App. F. Admitted rows are not averaged into a fixed network; each is an update operator that re-solves the quadratic program against the existing polytope. A row consistent with the rest of the network shrinks the polytope and lowers the median window; a conflicting row forces the feasibility factor c wider and surfaces in the widest-window list (§4) together with the specific edges it disagrees with. This yields a principled acceptance criterion for crowdsourced evidence (accept if the median window decreases, flag for adjudication if it increases, returning edge-specific feedback) and a continual-improvement loop: the collective polytope width is a monotone-decreasing measure of residual epistemic uncertainty that the system drives down both by adding consistent literature and by the structural changes its widest windows nominate. Because every update re-solves rather than overwrites, the window remains an accurate report of what is unresolved at each point in the network’s life, which a lifelong-learning deployment requires.